EFY TESTED
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Part 1

ANTI-COLLISION SYSTEM For Vehicles

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High-end cars now come with auto-braking, obstacle detection, and proximity alert, which are important parts of an Advanced Driver Assist System (ADAS). These use proximity sensors, cameras, and/or a lidar system. This project proposes to build a simple anti-collision system with a lidar.

Fig. 1 shows the lidar mounted on a car, along with the data generated by it for giving an alert. The components required to build the project are mentioned under Table 1.

You can use any lidar that is compatible with Python and has a Python SDK module for development, such as RPi lidar, YDLIDAR, Intel Real Sense lidar, and Velodyne lidar. The price of lidars available in the market varies considerably and so does their detection range. Most of them use USB serial communication.

In our design, we used YDLIDAR X2, but you can use any of the above-mentioned lidars. However, it is important to go through the datasheet (can search on Google) of the lidar and the Python module you are using to create the code around them.

In the datasheet, we can see the table that tells us that the field of view of this lidar is 360 degrees. So, this lidar can sense in any direction as it uses a motor to rotate. Some lidars provide 250- or 180-degree coverage only, so you need to choose the lidar as per your need.

The second important aspect of a lidar is its range and the conditions under which it can work. Some lidars use IR laser or red laser to work, some use doppler waves, and some others may be using yet another technology. So, to design a

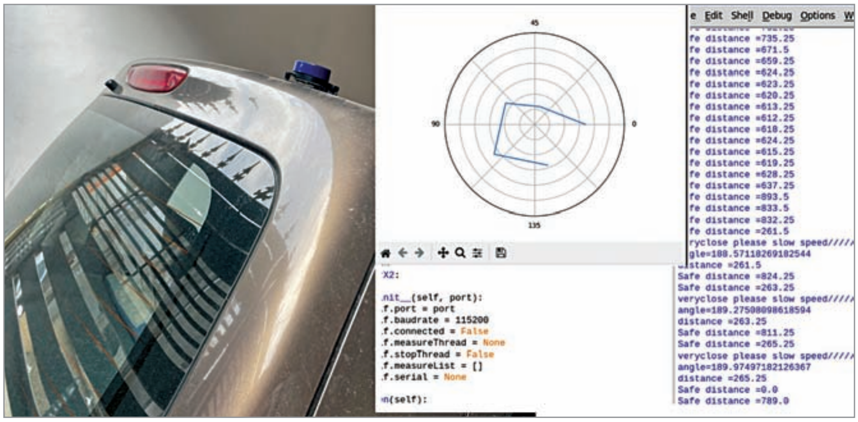

Fig. 1: Lidar system on a car and the data for alert

Fig. 2: YDLIDAR X2 (Credit: https://www.ydlidar.com)

TABLE 1
BILL OF MATERIAL

Components	Quantity	Description
360° laser lidar	1	YDLIDAR X2
Raspberry Pi / Nvidia Jetson	1	1GB - 2G
5V DC power supply	1	5V, 2A
USB cable	2	Type C to USB

TABLE 2
YDLIDAR X2 PARAMETERS

Item	Minimum	Typical	Maximum	Unit	Remarks
Ranging frequency	/	3000	/	Hz	Ranging 3000 times per second
Motor frequency	5	6	8	Hz	Need to connect to PWM signal, recommended to use the speed of 6Hz
Ranging distance	0.12	/	8	m	Indoor environment with 80% reflectivity
Field of view	/	0-360	/	Deg	/
Systematic error	/	2	/	cm	Range≤1m
Relative error	/	3.5%	/	/	1m<Range≤6m
Tilt angle	0.25	1	1.75	Deg	/
Angle resolution	0.60 (frequency @5Hz)	0.72 (frequency @6Hz)	0.96 (frequency @8Hz)	Deg	Different motor frequency

Note: It is factor FQC standard value, 80% reflectivity material object (Credit: https://www.ydlidar.com/Public)